

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Dushimimana Sandrine	Team member	s.dushimima2@alustudent.com				
IGIHOZO Nyituriki Marie Colombe	Team member	c.nyituriki@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Found and built the dataset using African health data	Colombe	03/06/2026	Completed	Yes	Dataset ready with missing values and non-numeric columns
	Handled the non-numeric columns and filled in the missing values	Sandrine	05/06/2026	Completed	Yes	Encoded Region column, dropped Country, filled NaN with column means
	Standardised the data so all features are on the same scale	Sandrine	07/06/2026	Completed	Yes	Checked that mean and std came out correct after standardising
	Computed the covariance matrix from the standardised data	Sandrine	07/06/2026	Completed	Yes	Matrix looked correct, features that should correlate did
	Ran eigendecomposition, sorted the components, and projected the data	Sandrine	07/06/2026	Completed	Yes	PC1 captured the most variance as expected
	Made the before and after PCA scatter plots	Colombe	07/06/2026	Completed	Yes	Both plots labelled correctly, added country annotations
	Chose how many components to keep and wrote the explanation	Colombe	09/06/2026	Completed	Yes	Went with 90% threshold and explained what gets lost
	Tested performance on bigger datasets and handled large data	Colombe	09/06/2026	Completed	Yes	Ran benchmarks, added chunked approach for memory efficiency
	Final checks, merged PDFs and uploaded everything to GitHub	Colombe & Sandrine	12/06/2026	Completed	Yes	Submission ready, all outputs visible in the notebook
Meeting Attendance Log						
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
03/06/2026	understand assignment requirements and plan work	Dushimimana Sandrine	Dushimimana Sandrine IGIHOZO Nyituriki Marie Colombe	Read assignment rubric together. Discussed PCA theory (eigenvalues, eigenvectors, covariance). Agreed on African Health & Disease Burden as dataset theme. Confirmed no sklearn allowed.	Colombe to research and design the dataset. Sandrine to begin reviewing PCA math and standardisation formula.	
06/06/2026	Dataset review and Task 1 planning: validate dataset and assign coding tasks	IGIHOZO Nyituriki Marie Colombe	Dushimimana Sandrine IGIHOZO Nyituriki Marie Colombe	Reviewed final dataset (54 African countries, 12 columns, NaN values confirmed). Discussed non-numeric handling: drop Country, encode Region. Allocated Task 1 coding to Sandrine.	Sandrine to implement standardisation, covariance, eigendecomposition, and projection. Colombe to implement visualisation (before/after PCA plots).	

09/06/2026	Task 1 progress check: review PCA implementation and visualisations	Dushimimana Sandrine	Dushimimana Sandrine IGIHOZO Nyituriki Marie Colombe	Sandrine demonstrated working PCA steps (standardisation verified: mean $\approx$ 0, std $\approx$ 1; covariance matrix correct; eigendecomposition sorted). Colombe showed before/after scatter plots. Minor axis label fixes agreed.	Sandrine to finalise projection and Step 7 output. Colombe to begin Task 2 (dynamic selection, written justification, loadings heatmap).	
11/06/2026	Tasks 2 & 3 completion: dynamic selection justification, benchmarking, final notebook review	IGIHOZO Nyituriki Marie Colombe	Dushimimana Sandrine IGIHOZO Nyituriki Marie Colombe	Colombe presented Task 2: 90% variance threshold selected, written justification covering economic activity and population pressure information loss, loadings heatmap completed. Sandrine reviewed Task 3 benchmark results (all under 100 ms for 50k rows). Both verified all notebook cell outputs are visible.	Both members to do final proofread. Sandrine to export notebook PDF. Colombe to export task sheet PDF. Merge both into one PDF and upload to GitHub by 12/06/2026.	
12/06/2026	Submission preparation: merge PDFs, GitHub upload, final checks	Dushimimana Sandrine	Dushimimana Sandrine IGIHOZO Nyituriki Marie Colombe	Final notebook PDF and task sheet PDF merged into one submission file. GitHub repository reviewed ; notebook, PDF, and task sheet all present. Confirmed submission link ready for upload on 13/06/2026.	Submit GitHub link via the course portal	